DESIGN COMPUTED · FREE GUIDE

A gentle slope carries far: how the Pont du Gard moved water 50 km on a one-inch fall

The free guide to the reel on the Pont du Gard: the numbers behind the Nimes aqueduct, why a tiny gradient works, how the blocks were lifted without mortar, and a bottle-and-tray experiment that shows how slope controls speed.



The bridge and the aqueduct

The Pont du Gard is 48.8 m high with three tiers of arches (6, 11 and 35 arches). It carried the aqueduct that brought spring water to the Roman city of Nemausus (Nimes) for fountains, baths and homes. Construction probably ended around 40 to 60 AD and took about fifteen years with 800 to 1,000 workers.

The channel runs about 50 km, although the straight-line distance is only about 20 km, because tunnelling through the Garrigues hills was judged impractical and the route winds around them.

The slope

The source at Fontaine d'Eure is 76 m above sea level, only 17 m higher than the basin in Nimes: an average gradient of about 1 in 3,000. On the bridge itself the channel descends 2.5 cm over 456 m, a gradient of 1 in 18,241.

That is so gentle that the water flows slowly and steadily: an estimated 40,000 cubic metres a day, taking nearly 27 hours to reach the city.

Built without mortar

The blocks were cut to fit by friction and gravity, and the bridge was built largely without mortar or clamps. About 50,400 tons of limestone (some blocks up to 6 tons) were lifted with cranes, block-and-tackle pulleys and a large human-powered treadwheel for the biggest blocks. Protruding blocks supported the scaffolding.

Built examples

Pont du Gard, France	48.8 m high, three tiers; UNESCO World Heritage since 1985.
Nimes aqueduct	About 50 km long with an average fall of 1 in 3,000.
Modern gravity sewers	Pipes laid at a few millimetres per metre use the same idea.

Try it yourself: slope and speed

1. Take a long tray or a length of gutter and a bottle of water with a small hole in the cap.
2. Prop one end 1 cm higher than the other and let a thin stream run down: time how long it takes to cross the tray.
3. Raise the end to 3 cm and repeat: the water is faster, but is it better controlled?
4. Now lower it to a few millimetres and watch the stream become slow and steady.
5. Find the smallest slope at which the water still moves all the way. That is how Roman engineers set their channel.

The rule to remember

If you let gravity do the work, a small constant slope beats a steep one over a long distance.

Default answer	Add height, pumps or a steeper drop.
Form-finder answer	Hold a tiny gradient for the whole length.
Check	Ask: what slope would my design need if I could not use a pump?

Go further

Pont du Gard (Wikipedia)
https://en.wikipedia.org/wiki/Pont_du_Gard

Nimes aqueduct (Wikipedia)
https://en.wikipedia.org/wiki/Nemausus_aqueduct

UNESCO: Pont du Gard
https://whc.unesco.org/en/list/344

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Facts: Wikipedia (Pont du Gard), UNESCO.

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